3 A rock climber, of total mass 62 kg, holds herself in horizontal equilibrium against a vertical cliff. She pulls on a rope that is fixed at the top of the cliff and presses her feet against the cliff.

Fig. 3.1 shows her position.

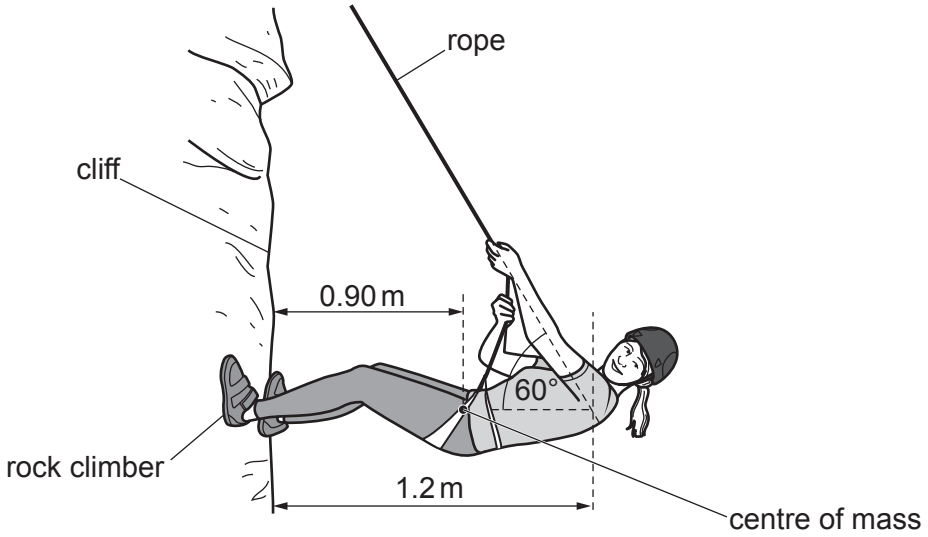


Fig. 3.1 (not to scale)

(a) Calculate the total weight of the climber.

weight = ... [1]

(b) State the **two** conditions needed for equilibrium.

1. .....

2. .....

[2]

(c) The climber's centre of mass is 0.90 m from the cliff.

(i) Calculate the moment about her feet due to her weight.

moment = ... [2]

(ii) The line of the rope meets the horizontal line through her centre of mass at a distance of 1.2 m from the cliff, as shown in Fig. 3.1. The rope is at an angle of 60° to the horizontal.

Determine the tension in the rope.

tension = ... [3]

[Total: 8]